EXPERIMENT NO : 4

STUDY OF LAN CONFIGURATION

AIM

To establish a Local Area Network (LAN) by interconnecting multiple computers through an Ethernet switch and to verify communication and resource sharing among the connected systems.

OBJECTIVE

- To understand the basic components of a LAN.
- To configure IPv4 addresses on computers.
- To connect multiple systems using an Ethernet switch.
- To test network connectivity using the ping command.
- To demonstrate file sharing between computers on the same network.

REQUIREMENTS

- 4 Computers
- 1 Ethernet Switch
- 4 Ethernet/Cat6 Cables
- Windows Operating System
- Command Prompt

NETWORK TOPOLOGY

A simple **star topology** is used, where all four computers are connected to a central Ethernet switch.

Example IP Configuration:

Device	IP Address	Subnet Mask
PC1	192.168.20.1	255.255.255.0
PC2	192.168.20.2	255.255.255.0
PC3	192.168.20.3	255.255.255.0
PC4	192.168.20.4	255.255.255.0

PROCEDURE

1. Determine the number of computers and network devices required for the LAN.
2. Arrange the computers and switch according to the planned network layout.
3. Connect each computer to a separate port on the Ethernet switch using Ethernet cables.
4. Check the link indicators on the switch and computers to confirm that the physical connections are active.
5. Configure the IPv4 address on each computer:
    - Press **Windows + R** to open the Run dialog.
    - Enter ncpa.cpl and press Enter.
    - Right-click the **Ethernet** adapter and select **Properties**.
    - Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**.
    - Choose **Use the following IP address**.
    - Enter a unique IP address and a common subnet mask for all computers.
6. Repeat the IP configuration process for the remaining PCs, ensuring that no two computers have the same IP address.
7. If the switch supports web-based management, connect a computer to the switch and open the switch management interface using its configured IP address.
8. Log in with the appropriate administrator credentials and verify the switch's network settings.
9. Open **Command Prompt** on one computer.
10. Use the following command to test communication with another PC:

```
ping 192.168.20.2
```

11. Check the responses received from the destination computer.
12. Repeat the ping test between different pairs of computers.
13. Create a test folder on one computer and enable **Network Sharing** for that folder.
14. From another computer connected to the same LAN, open the network location and access the shared folder.
15. Verify that files can be accessed through the LAN.

OBSERVATION

All four computers are connected to a central switch, forming a star-based LAN. Each system is assigned a unique IP address within the same subnet. Successful ping responses indicate that the computers can communicate with one another through the switch.

The shared-folder test demonstrates that computers connected to the same LAN can exchange resources when appropriate sharing and permission settings are enabled.

RESULT

Thus, a Local Area Network was successfully configured using multiple computers, Ethernet cables, and a network switch. IP addressing, connectivity testing using the ping command, and file sharing between LAN-connected computers were successfully verified.

CONCLUSION

The experiment demonstrated the basic working of a LAN and the role of a switch in connecting multiple devices. It also provided practical knowledge of IPv4 configuration, network connectivity testing, and resource sharing within a local network.